MTH202 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

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MTH202 (Lectures 23–45)

Mathematical Statement MCQs (1–50)

Q1. The statement (1+2+\cdots+n=\frac{n(n+1)}{2}) is true for all positive integers.
A) Only for even n
B) Only for odd n
C) For every positive integer ✅
D) Only for prime n

Q2. In mathematical induction, the first step is called:
A) Recursive Step
B) Basis Step ✅
C) Final Step
D) Expansion Step

Q3. If (P(k)) is assumed true to prove (P(k+1)), this is known as:
A) Contradiction
B) Basis Step
C) Inductive Hypothesis ✅
D) Converse

Q4. The sum (1+3+5+\cdots+(2n-1)) equals:
A) (2n)
B) (n(n+1))
C) (n^2) ✅
D) (2n^2)

Q5. The sum (1+2+2^2+\cdots+2^n) equals:
A) (2^n-1)
B) (2^{n+1})
C) (2^{n+1}-1) ✅
D) (2^n+1)

Q6. The formula (\sum_{i=1}^{n}i^2=\frac{n(n+1)(2n+1)}6) represents:
A) Sum of cubes
B) Sum of squares ✅
C) Arithmetic series
D) Geometric series

Q7. The telescoping series
(\frac1{1\cdot2}+\frac1{2\cdot3}+\cdots+\frac1{n(n+1)}) equals:
A) (\frac n2)
B) (\frac n{n+1}) ✅
C) (n)
D) (\frac1n)

Q8. For (n=3), (1+2+3=)
A) 5
B) 6 ✅
C) 7
D) 9

Q9. For (n=4), (1+3+5+7=)
A) 12

B) 14
C) 16 ✓
D) 18

Q10. (1+2+4+8+16=)
A) 30
B) 31 ✓
C) 32
D) 33

Q11. (1^2+2^2+3^2=)
A) 12
B) 13
C) 14 ✓
D) 15

Q12. The Basis Step verifies:
A) (P(0)) or (P(1)) depending on the statement ✓
B) (P(k+1))
C) (P(n+2))
D) None

Q13. In induction, after assuming (P(k)), we prove:
A) (P(k-1))
B) (P(2k))
C) (P(k+1)) ✓
D) (P(k+2))

Q14. If (d|n), then:
A) (n=d+k)
B) (n=dk) for some integer (k) ✓
C) (d=n+k)
D) None

Q15. (3|18) means:
A) 18 divides 3
B) 3 divides 18 ✓
C) 3 is greater than 18
D) None

Q16. (18) is a multiple of:
A) 5
B) 4
C) 3 ✓
D) 7

Q17. (2^{2(2)}-1=)
A) 12
B) 14
C) 15 ✓
D) 17

Q18. (15) is divisible by:
A) 4
B) 5

C) 3
D) Both B and C ✅

Q19. (2^6-1=)
A) 61
B) 63 ✅
C) 65
D) 67

Q20. (3^3-3=)
A) 18
B) 21
C) 24 ✅
D) 30

Q21. (24) is divisible by:
A) 4
B) 6
C) Both A and B ✅
D) 5

Q22. Product of two consecutive integers is always divisible by:
A) 3
B) 2 ✅
C) 5
D) 7

Q23. (7\times8) is divisible by:
A) 2 ✅
B) 5
C) 9
D) 11

Q24. (5^3-5=)
A) 115
B) 120 ✅
C) 125
D) 130

Q25. (5^3-5) is divisible by:
A) 2
B) 3
C) 6 ✅
D) 8

Q26. (x^n-y^n) is divisible by:
A) (x+y)
B) (x-y) ✅
C) (xy)
D) (x^2-y^2)

Q27. If (a|b) and (b|c), then:
A) (a|c) ✅
B) (c|a)

C) (a+b=c)
D) None

Q28. (2n+1<2^n) holds for:
A) (n\ge3) ✓
B) (n\ge1)
C) (n\ge2)
D) Every integer

Q29. For (n=4), (2n+1=)
A) 7
B) 8
C) 9 ✓
D) 10

Q30. (2^4=)
A) 8
B) 12
C) 16 ✓
D) 18

Q31. The sequence (a_1=2,\ a_n=5a_{n-1}) satisfies:
A) (a_n=2\cdot5^{n-1}) ✓
B) (5^n)
C) (2n)
D) (5n)

Q32. (a_4) equals:
A) 125
B) 200
C) 250 ✓
D) 300

Q33. If (d_1=2,\ d_n=\frac{d_{n-1}}n), then:
A) (d_n=\frac2{n!}) ✓
B) (2n!)
C) (n!)
D) (2^n)

Q34. Direct proof starts by assuming:
A) Conclusion is false
B) Premise is true ✓
C) Both false
D) None

Q35. Proof by contradiction starts by assuming:
A) Statement is true
B) Statement is false ✓
C) Both true
D) Nothing

Q36. Contrapositive of (p\rightarrow q) is:
A) (q\rightarrow p)
B) (\neg p\rightarrow\neg q)

C) (\neg q\rightarrow\neg p) ✓
D) (q\rightarrow\neg p)

Q37. Every even integer has form:
A) (2k) ✓
B) (2k+1)
C) (k+2)
D) (3k)

Q38. Every odd integer has form:
A) (2k)
B) (2k+1) ✓
C) (3k)
D) (4k)

Q39. Sum of two odd integers is:
A) Odd
B) Even ✓
C) Prime
D) Irrational

Q40. Product of an even and an odd integer is:
A) Odd
B) Prime
C) Even ✓
D) Composite

Q41. Square of an even integer is:
A) Odd
B) Even ✓
C) Prime
D) Irrational

Q42. If (n) is odd, then (n^3+n) is:
A) Odd
B) Even ✓
C) Prime
D) Composite

Q43. Sum of two rational numbers is:
A) Irrational
B) Rational ✓
C) Prime
D) Integer only

Q44. If (a|b) and (a|c), then:
A) (a|(b+c)) ✓
B) (b|c)
C) (c|a)
D) None

Q45. Sum of three consecutive integers is divisible by:
A) 2
B) 3 ✓

C) 4
D) 5

Q46. Counterexample is used to:
A) Prove every statement
B) Disprove a false statement ✅
C) Perform induction
D) Solve equations

Q47. (n^2-n+11) is not prime for:
A) (n=5)
B) (n=11) ✅
C) (n=3)
D) (n=2)

Q48. A counterexample for "If (a<b), then (a^2<b^2)" is:
A) (a=1,b=2)
B) (a=-5,b=-2) ✅
C) (a=2,b=3)
D) (a=3,b=5)

Q49. There is no greatest integer because:
A) Integers end at infinity
B) (N+1>N) for every integer (N) ✅
C) 1 is greatest
D) None

Q50. If (n^2) is even, then (n) is:
A) Odd
B) Even ✅
C) Prime
D) Composite

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Mathematical Statement MCQs (51–100)

Q51. If (2^{2n}-1) is considered in induction, then it is divisible by:
A) 2
B) 3 ✅
C) 5
D) 7

Q52. For (n=3), (2^{2n}-1=)
A) 61
B) 62
C) 63 ✅
D) 64

Q53. (63) is divisible by:
A) 4
B) 5
C) 3 ✅
D) 8

Q54. (6^3-6=)
A) 208
B) 210 ✅
C) 212
D) 214

Q55. (210) is divisible by:
A) 5
B) 6
C) Both A and B ✅
D) 8

Q56. (x^5-y^5) is always divisible by:
A) (x+y)
B) (x-y) ✅
C) (xy)
D) (x^2-y^2)

Q57. If (4|20), then:
A) 20 is a divisor of 4
B) 4 divides 20 ✅
C) 4 is greater than 20
D) None

Q58. If (5|35), then 35 can be written as:
A) (5+7)
B) (5\times7) ✅
C) (7-5)
D) (5^7)

Q59. The product (12\times13) is divisible by:
A) 2 ✅
B) 5
C) 7
D) 9

Q60. (9^3-9=)
A) 720 ✅
B) 721
C) 722
D) 723

Q61. (720) is divisible by:
A) 6 ✅
B) 7
C) 11
D) 13

Q62. If (a|b) and (b|c), then:
A) (c|a)
B) (a|c) ✅
C) (a=c)
D) None

Q63. (2(5)+1=)
A) 9
B) 10
C) 11 ✓
D) 12

Q64. (2^5=)
A) 16
B) 24
C) 32 ✓
D) 64

Q65. The inequality (2n+1<2^n) first becomes true at:
A) (n=1)
B) (n=2)
C) (n=3) ✓
D) (n=4)

Q66. If (a_1=2,;a_n=5a_{n-1}), then (a_5=)
A) 1250 ✓
B) 625
C) 2500
D) 500

Q67. (2\cdot5^4=)
A) 625
B) 1250 ✓
C) 250
D) 3125

Q68. (d_n=\frac2{n!}). Then (d_4=)
A) (\frac1{12}) ✓
B) (\frac16)
C) (\frac18)
D) (\frac1{24})

Q69. (4!=)
A) 12
B) 20
C) 24 ✓
D) 48

Q70. The proof that starts by assuming the hypothesis true is:
A) Direct Proof ✓
B) Contradiction
C) Contrapositive
D) Counterexample

Q71. Assuming the statement is false is the basis of:
A) Direct Proof
B) Contradiction ✓
C) Converse
D) Deduction

Q72. Contrapositive of (p\rightarrow q) is:
A) (q\rightarrow p)
B) (\neg q\rightarrow\neg p) ✅
C) (\neg p\rightarrow\neg q)
D) (q\rightarrow\neg p)

Q73. Every even integer is expressible as:
A) (2k) ✅
B) (2k+1)
C) (3k)
D) (4k+1)

Q74. Every odd integer is:
A) (2k)
B) (2k+1) ✅
C) (3k)
D) (5k)

Q75. Sum of an odd and an even integer is:
A) Even
B) Odd ✅
C) Prime
D) Composite

Q76. Product of two odd integers is:
A) Even
B) Odd ✅
C) Prime
D) Rational

Q77. Square of an odd integer is:
A) Even
B) Odd ✅
C) Prime
D) Zero

Q78. If (n) is even, then ((-1)^n=)
A) -1
B) 0
C) 1 ✅
D) 2

Q79. ((-1)^8=)
A) -1
B) 0
C) 1 ✅
D) 8

Q80. If (n) is odd, then (n^3+n) is:
A) Even ✅
B) Odd
C) Prime
D) Composite

Q81. Sum of two rational numbers is:
A) Irrational
B) Rational ✅
C) Integer only
D) Prime

Q82. (\frac34+\frac14=)
A) (\frac12)
B) 1 ✅
C) (\frac54)
D) 2

Q83. If (a|b) and (a|c), then:
A) (a|(b+c)) ✅
B) (b|a)
C) (c|a)
D) None

Q84. If (4|8) and (4|12), then (4|)
A) 18
B) 20 ✅
C) 22
D) 26

Q85. Three consecutive integers always have sum divisible by:
A) 2
B) 3 ✅
C) 4
D) 5

Q86. (8+9+10=)
A) 25
B) 26
C) 27 ✅
D) 28

Q87. (27) is divisible by:
A) 2
B) 3 ✅
C) 4
D) 5

Q88. Counterexample is used to:
A) Verify
B) Disprove ✅
C) Expand
D) Integrate

Q89. The polynomial (n^2-n+11) fails to be prime at:
A) (n=10)
B) (n=11) ✅
C) (n=9)
D) (n=8)

Q90. ($11^2-11+11=$)
A) 110
B) 121 ✅
C) 132
D) 111

Q91. (121) is:
A) Prime
B) Composite ✅
C) Irrational
D) Even

Q92. A valid counterexample to ($a<b\Rightarrow a^2<b^2$) is:
A) 2,3
B) -5,-2 ✅
C) 1,4
D) 4,5

Q93. The contradiction "There is no greatest integer" is obtained by considering:
A) (N-1)
B) (N+1) ✅
C) (2N)
D) (-N)

Q94. If (n^2) is even, then (n) is:
A) Odd
B) Even ✅
C) Prime
D) Composite

Q95. The statement "If (n) is divisible by 5 then (n^2) is divisible by 25" is proved using:
A) Counterexample
B) Contrapositive ✅
C) Converse
D) Induction only

Q96. If ($n=15$), then ($n^2=$)
A) 215
B) 225 ✅
C) 250
D) 275

Q97. (225) is divisible by:
A) 5
B) 25
C) Both A and B ✅
D) 15 only

Q98. If ($|x|>1$), then:
A) ($x>1$) or ($x<-1$) ✅
B) ($x>0$)
C) ($x<1$)
D) ($x=1$)

Q99. The implication $(p\rightarrow q)$ is logically equivalent to:
A) Its converse
B) Its contrapositive ✅
C) Its inverse
D) None

Q100. Mathematical induction is mainly used to prove statements involving:
A) Real numbers only
B) Positive integers ✅
C) Complex numbers only
D) Matrices only

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Difficult Mathematical Statement MCQs (101–125)

Q101. Assume $(P(k):1+2+\cdots+k=\frac{k(k+1)}2)$. During the inductive step, the left-hand side of $(P(k+1))$ is correctly written as:
A) $(\frac{k(k+1)}2+k)$
B) $(\frac{k(k+1)}2+(k+1))$ ✅
C) $(\frac{k(k+1)}2+1)$
D) $(\frac{(k+1)(k+2)}2)$

Q102. If $(a_n=2\cdot5^{n-1})$, then the ratio $(\frac{a_{n+1}}{a_n})$ is:
A) 2
B) 4
C) 5 ✅
D) (5^n)

Q103. The recursive sequence
$(a_1=2,;a_n=5a_{n-1})$
has $(a_6=)$
A) 3125
B) 6250 ✅
C) 12500
D) 15625

Q104. If $(d_n=\frac2{n!})$, then
$(\frac{d_{n+1}}{d_n})$ equals:
A) (n)
B) $(n+1)$
C) $(\frac1{n+1})$ ✅
D) $(2(n+1))$

Q105. Which expression is always an integer?
A) $(\frac{n}{2})$

B) (\frac{n(n+1)}2) ✅
C) (\frac{n+2}{3})
D) (\frac{2n+1}{2})

Q106. During induction, multiplying both sides of
(1+kx\le(1+x)^k)
by ((1+x)) is valid because:
A) (x>0)
B) (x\ge0)
C) (1+x>0) ✅
D) (k>0)

Q107. In proving
(1+(k+1)x\le(1+x)^{k+1}),
the non-negative term responsible for the inequality is:
A) (kx)
B) (k+x)
C) (kx^2) ✅
D) (x+k^2)

Q108. If (a|b) and (b|c), then writing
(b=ar,;c=bs)
implies
(c=)
A) (a+r+s)
B) (ars) ✅
C) (a(r+s))
D) (ab+s)

Q109. Which statement is **always true**?
A) Every prime number is odd.
B) Every composite number is even.
C) Every even integer can be written as (2k). ✅
D) Every odd integer is prime.

Q110. If (m+n) is even, then:
A) Both integers must be even.
B) Both integers must be odd.
C) They have the same parity. ✅
D) One is even and one is odd.

Q111. Which pair contradicts the statement
"if (a<b), then (a^2<b^2)"?
A) (2,3)
B) (-3,-2) ✅
C) (1,5)
D) (0,4)

Q112. If (n=2k+1), then
(n^2=)
A) (4k^2+1)
B) (4k^2+4k+1) ✅
C) (2k^2+1)
D) (4k+1)

Q113. The expression
(n^3+n)
is always even whenever:
A) (n) is even only
B) (n) is odd only ✅
C) (n) is prime
D) (n) is composite

Q114. Which implication is proved most naturally by contradiction?
A) Sum of rationals is rational.
B) Product of evens is even.
C) If (n^2) is even, then (n) is even. ✅
D) Sum of consecutive integers.

Q115. In the proof that (\sqrt2) is irrational, the contradiction arises because:
A) (m=n)
B) Both (m) and (n) become even. ✅
C) (m<n)
D) (n=1)

Q116. If (n^2) is divisible by (25), then:
A) (n) must be divisible by 5. ✅
B) (n) is prime.
C) (n) is odd.
D) (n) is composite.

Q117. Which statement is FALSE?
A) Every prime greater than 2 is odd.

B) Every odd integer has the form (2k+1).
C) Every irrational number squared is irrational. ✅
D) Every even integer has the form (2k).

Q118. If
(r=\frac ab,;s=\frac cd),
then
(r+s=)
A) (\frac{a+c}{b+d})
B) (\frac{ad+bc}{bd}) ✅
C) (\frac{ab+cd}{bd})
D) (\frac{a+b}{c+d})

Q119. Which statement requires only **one counterexample** to disprove?
A) Every odd number is prime. ✅
B) Every integer equals itself.
C) Every square is non-negative.
D) Sum of integers is integer.

Q120. Which expression proves three consecutive integers are divisible by 3?
A) (3n+2)
B) (3(n+1)) ✅
C) (n(n+1))
D) (2(n+1))

Q121. If (a|b) and (a|c), then (a|(b-c)) because:
A) (b-c=a(r-s)) ✅
B) (b-c=a+r-s)
C) (b-c=rs)
D) None

Q122. The contrapositive of
"If (n^2) is even, then (n) is even"
is:
A) If (n) is even, then (n^2) is even.
B) If (n) is odd, then (n^2) is odd. ✅
C) If (n^2) is odd, then (n) is even.
D) If (n) is odd, then (n^2) is even.

Q123. Which statement is logically equivalent to
(p\rightarrow q)?
A) (q\rightarrow p)

B) $(\neg p\rightarrow\neg q)$
C) $(\neg q\rightarrow\neg p)$ ✅
D) $(q\lor p)$

Q124. The proof that there is **no greatest integer** depends on the fact that:
A) $(N-1)$ exists.
B) $(N+1)$ is also an integer. ✅
C) Zero exists.
D) Integers are finite.

Q125. In mathematical induction, the inductive hypothesis is used **only** to:
A) Assume the statement false.
B) Verify the base case.
C) Prove the statement for the next integer $(k+1)$. ✅
D) Prove the converse.

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Difficult Mathematical Statement MCQs (126–150)

Q126. Let $(a_1=2)$ and $(a_n=5a_{n-1})$. Which statement correctly represents the closed form of the sequence?
A) $(a_n=5^n)$
B) $(a_n=2\cdot5^{,n-1})$ ✅
C) $(a_n=2^n)$
D) $(a_n=5n)$

Q127. If $(a_n=2\cdot5^{n-1})$, then $(\dfrac{a_8}{a_5})$ is equal to:
A) 25
B) 75
C) 125 ✅
D) 625

Q128. Suppose $(d_1=2)$ and $(d_n=\dfrac{d_{n-1}}{n})$. Then (d_6) equals:
A) $(\dfrac1{180})$ ✅
B) $(\dfrac1{120})$
C) $(\dfrac1{240})$
D) $(\dfrac1{360})$

Q129. If $(P(k))$ is true and
$$P(k+1)=P(k)+(k+1),$$
then after substitution the expression becomes:

A) (\dfrac{k(k+1)}2+k)
B) (\dfrac{k(k+1)}2+(k+1)) ✅
C) (\dfrac{k(k-1)}2+(k+1))
D) (\dfrac{k(k+2)}2)

Q130. Which statement is **essential** for proving
[
1+(k+1)x\le(1+x)^{k+1}
]
by induction?
A) (k<0)
B) (x^2\ge0) ✅
C) (x=-1)
D) (k=1)

Q131. If (n=2k+1), then
[
n^3=(2k+1)^3
]
contains how many odd-power terms of (k)?
A) One
B) Two ✅
C) Three
D) Four

Q132. If (m+n) is even and (m) is odd, then (n) must be:
A) Even
B) Odd ✅
C) Prime
D) Composite

Q133. Which statement is logically equivalent to proving
[
p\rightarrow q
]
by contradiction?
A) Assume (p\land\neg q). ✅
B) Assume (\neg p\land q).
C) Assume (q\rightarrow p).
D) Assume (\neg p\rightarrow\neg q).

Q134. The contradiction in the proof that (\sqrt2) is irrational is obtained because:
A) (m=n)
B) Both numerator and denominator become divisible by 2. ✅

C) (m<n)
D) (n=1)

Q135. Which statement is TRUE for every integer (n)?
A) (n^2+n) is odd.
B) (n(n+1)) is even. ✅
C) (n^2-1) is prime.
D) (2n+1) is even.

Q136. If (a|b) and (a|(b+c)), then (a|c) because:
A) (c=(b+c)-b) ✅
B) (c=b+(b+c))
C) (c=ab)
D) None

Q137. Which expression is always divisible by 6?
A) (n^2+n)
B) (n^3-n) ✅
C) (n^2+1)
D) (2n+3)

Q138. If (n=9), then (n^3-n) equals:
A) 702
B) 720 ✅
C) 729
D) 738

Q139. Which statement is FALSE?
A) Every rational number is real.
B) Every irrational number is real.
C) Every real number is rational. ✅
D) Sum of rationals is rational.

Q140. A direct proof usually begins with:
A) Assuming the conclusion.
B) Assuming the hypothesis. ✅
C) Assuming the negation.
D) Assuming the converse.

Q141. Which of the following is a valid counterexample to
"Every prime number (p) has (p+2) prime"?
A) (p=3)
B) (p=5)
C) (p=7) ✅
D) (p=11)

Q142. If (a|b), then (a|(5b)) because:
A) (5b=a(5k)) ✅
B) (5b=a+k)
C) (5b=ak+5)
D) None

Q143. Which number satisfies
[
2^n-1
]
being prime?
A) (n=4)
B) (n=6)
C) (n=7) ✅
D) (n=8)

Q144. Which implication is proved most naturally by contraposition?
A) If (n^2) is even, then (n) is even. ✅
B) Sum of rationals is rational.
C) Three consecutive integers are divisible by 3.
D) Product of evens is even.

Q145. If (|x|>1), then:
A) (x\neq0)
B) (x>1) or (x<-1) ✅
C) (x>0)
D) (x<0)

Q146. Which statement is always TRUE?
A) Product of two irrational numbers is irrational.
B) Product of two irrational numbers may be rational. ✅
C) Sum of two irrational numbers is always irrational.
D) Square root of every integer is irrational.

Q147. If (p) is prime and (p|a) as well as (p|(a+1)), then:
A) (p=2)
B) (p|1), giving a contradiction. ✅
C) (a=1)
D) (p=1)

Q148. The proof that there are infinitely many prime numbers constructs:
A) A smaller prime.
B) (N=p_1p_2\cdots p_n+1). ✅
C) (N=p_1+p_2+\cdots+p_n).
D) (N=2^n-1).

Q149. Which statement is TRUE regarding mathematical induction?
A) It proves only inequalities.
B) It proves statements indexed by positive integers. ✅
C) It is used only for recursive sequences.
D) It replaces direct proof.

Q150. Which of the following requires the **strongest understanding of mathematical induction**?
A) Verifying only the basis step.
B) Assuming (P(k)) and correctly transforming algebraic expressions to establish (P(k+1)). ✅
C) Memorizing recursive formulas.
D) Checking numerical values for several integers only.

MTH202 (Lectures 23–45)

Difficult Mathematical Statement MCQs (151–200)

Q151. If (P(1)) is true but the inductive step fails, then the statement is:
A) Proven true
B) Proven false
C) Not established by induction ✅
D) Always true

Q152. In mathematical induction, proving (P(k+1)) without using (P(k)) generally means:
A) The proof is invalid.
B) The proof may still be valid if (P(k+1)) is proved directly. ✅
C) The basis step is unnecessary.
D) The theorem is false.

Q153. The recursive sequence (a_n=5a_{n-1}) grows:
A) Linearly
B) Quadratically

C) Exponentially ✅
D) Logarithmically

Q154. If (a_n=2\cdot5^{n-1}), then (a_7-a_6=)
A) 6250
B) 5000 ✅
C) 2500
D) 1250

Q155. Which expression is always divisible by 2?
A) (n(n+1)) ✅
B) (n^2+1)
C) (2n+1)
D) (n^3+1)

Q156. If (n) is odd, then (n^2+2n) is:
A) Even
B) Odd ✅
C) Prime
D) Composite

Q157. Which statement is always true?
A) Product of two consecutive integers is even. ✅
B) Sum of two consecutive integers is even.
C) Difference of consecutive integers is even.
D) Product of consecutive integers is odd.

Q158. If (a|b), then (a|(7b)) because:
A) (7b=a(7k)) ✅
B) (7b=a+k)
C) (7b=7+a)
D) None

Q159. Which statement is FALSE?
A) Every prime number greater than 2 is odd.
B) Every even integer is composite. ✅
C) 2 is prime.
D) Every odd integer has form (2k+1).

Q160. If $(m+n)$ is odd, then:
A) Both are even.
B) Both are odd.
C) One is even and one is odd. ✓
D) Both are prime.

Q161. Which statement correctly describes proof by contradiction?
A) Assume the conclusion is true.
B) Assume the statement is false and derive a contradiction. ✓
C) Assume both hypothesis and conclusion are false.
D) Verify numerical examples only.

Q162. If $(\sqrt2=\frac{m}{n})$ is assumed in lowest terms, the contradiction is that:
A) $(m=n)$
B) Both (m) and (n) become divisible by 2. ✓
C) $(m<n)$
D) $(n=0)$

Q163. Which implication is logically equivalent to $(p\rightarrow q)$?
A) $(q\rightarrow p)$
B) $(\neg p\rightarrow\neg q)$
C) $(\neg q\rightarrow\neg p)$ ✓
D) $(q\lor p)$

Q164. If $(a|b)$ and $(a|c)$, then $(a|(5b-2c))$ because:
A) $(5b-2c=a(5r-2s))$ ✓
B) $(5b-2c=a+r+s)$
C) $(5b-2c=ab-c)$
D) None

Q165. Which number is a counterexample to the statement "Every odd number is prime"?
A) 9 ✓
B) 7
C) 11
D) 13

Q166. If $(n=2k)$, then (n^2) equals:
A) $(2k^2)$
B) $(4k^2)$ ✓
C) $(2k+1)$
D) $(4k+1)$

Q167. Which statement is TRUE?
A) Sum of two irrational numbers is always irrational.
B) Product of two irrational numbers is always irrational.
C) Sum of a rational and an irrational number is irrational. ✅
D) Every irrational number is prime.

Q168. The sequence (2,10,50,250,\ldots) is:
A) Arithmetic
B) Geometric with ratio 5 ✅
C) Harmonic
D) Constant

Q169. If (d_n=\frac2{n!}), then:
A) (d_{n+1}>d_n)
B) (d_{n+1}<d_n) ✅
C) Both equal
D) None

Q170. Which proof method is most suitable to disprove a universal statement?
A) Induction
B) Counterexample ✅
C) Contrapositive
D) Direct proof

Q171. Which expression is always divisible by 3?
A) (n+n+1+n+2) ✅
B) (n+n+1)
C) (2n+1)
D) (n^2+1)

Q172. If (a|b), then:
A) (a|b^2) ✅
B) (b|a^2)
C) (a+b=b)
D) None

Q173. Which statement is TRUE?
A) Every rational number is an integer.
B) Every integer is rational. ✅

C) Every irrational number is integer.
D) Every prime is irrational.

Q174. If (n) is even, then (3n+2) is:
A) Odd
B) Even ✅
C) Prime
D) Composite

Q175. The contrapositive of
"If (n) is divisible by 5, then (n^2) is divisible by 25"
is:
A) If (n^2) is not divisible by 25, then (n) is not divisible by 5. ✅
B) If (n) is not divisible by 5, then (n^2) is not divisible by 25.
C) If (n^2) is divisible by 25, then (n) is divisible by 5.
D) None

Q176. Which statement is FALSE?
A) Every composite number has more than two factors.
B) Every prime has exactly two positive divisors.
C) 1 is a prime number. ✅
D) 2 is the smallest prime.

Q177. If (P(k)) implies (P(k+1)) and (P(1)) is true, then:
A) (P(n)) is true for all positive integers. ✅
B) Only (P(2)) is true.
C) Only odd values satisfy (P(n)).
D) Nothing can be concluded.

Q178. Which expression simplifies to an even integer?
A) ((2k+1)+(2m+1)) ✅
B) ((2k)+(2m+1))
C) ((2k+1)+1)
D) (2k+3)

Q179. If (a|b) and (b=18), then which cannot divide 18?
A) 2
B) 3
C) 5 ✅
D) 6

Q180. Which proof technique assumes (n) is odd to prove "(n^2) is odd"?
A) Direct proof ✅
B) Contradiction
C) Counterexample
D) Mathematical induction

Q181. Which statement is always true?
A) Every square number is non-negative. ✅
B) Every square number is prime.
C) Every square number is odd.
D) Every square number is irrational.

Q182. If (n) is divisible by both 2 and 3, then (n) is divisible by:
A) 5
B) 6 ✅
C) 8
D) 9

Q183. If (n=11), then (n^2-n+11=)
A) 111
B) 121 ✅
C) 132
D) 143

Q184. Which theorem is commonly proved using contradiction?
A) Sum of evens is even.
B) (\sqrt2) is irrational. ✅
C) Product of evens is even.
D) Sum of rationals is rational.

Q185. Which statement is TRUE?
A) There exists a greatest integer.
B) There exists a smallest positive integer. ✅
C) There exists a largest prime.
D) There exists a largest natural number.

Q186. The proof of infinitely many primes starts by assuming:
A) There are infinitely many primes.
B) There are finitely many primes. ✅
C) Every odd number is prime.
D) Every prime is even.

Q187. The number
(N=p_1p_2\cdots p_n+1)
is constructed to obtain:
A) A contradiction ✅
B) An induction step
C) A recursive formula
D) A counterexample

Q188. Which is NOT a proof technique?
A) Contradiction
B) Contrapositive
C) Counterexample
D) Approximation ✅

Q189. Which statement is TRUE?
A) Direct proof starts with the negation.
B) Contradiction starts with the negation. ✅
C) Counterexample proves universal statements.
D) Induction disproves statements.

Q190. If (a|b), then (a|(kb)) for:
A) Positive integers only
B) Integers (k) ✅
C) Prime numbers only
D) Even numbers only

Q191. Which recursive sequence decreases?
A) (d_n=\frac{d_{n-1}}n) ✅
B) (a_n=5a_{n-1})
C) (a_n=a_{n-1}+5)
D) (a_n=2a_{n-1})

Q192. If (n) is odd, then (n^2) is:
A) Even
B) Odd ✅
C) Prime
D) Composite

Q193. Which statement is TRUE?
A) Every irrational number squared is irrational.

B) (($\sqrt2)^2$) is rational. ✅
C) Every irrational number is prime.
D) Every irrational number is negative.

Q194. If (m+n) is even and (m) is even, then (n) is:
A) Odd
B) Even ✅
C) Prime
D) Composite

Q195. Which theorem depends on parity arguments?
A) Sum of two odd integers is even. ✅
B) Every rational is irrational.
C) Every prime is composite.
D) None

Q196. Which statement is always FALSE?
A) Every even integer greater than 2 is prime. ✅
B) Every prime greater than 2 is odd.
C) Every integer is rational.
D) Sum of rationals is rational.

Q197. If (n^2) is odd, then (n) must be:
A) Even
B) Odd ✅
C) Prime
D) Composite

Q198. The strongest method for proving statements indexed by positive integers is:
A) Counterexample
B) Mathematical Induction ✅
C) Contradiction only
D) Contrapositive only

Q199. Which statement is TRUE?
A) Every universal statement is true.
B) One counterexample disproves a universal statement. ✅
C) One example proves a universal statement.
D) Counterexamples prove implications.

Q200. The **most essential requirement** for a valid mathematical induction proof is:
A) Checking many numerical cases.
B) Establishing both the **Basis Step** and the **Inductive Step** correctly. ✅
C) Using contradiction.
D) Using contraposition.

MTH202 Important Exam Points with Questions & Answers (Lectures 23–45)

Lecture 23 – Mathematical Induction

Important Exam Points

Mathematical induction is used to prove statements involving positive integers.

The two essential steps of induction are the Basis Step and the Inductive Step.

In the Basis Step, verify that the statement is true for the initial value (usually (n=1) or (n=2)).

In the Inductive Step, assume the statement is true for (n=k).

This assumption is called the Inductive Hypothesis.

Using the Inductive Hypothesis, prove that the statement is true for (n=k+1).

If both steps are completed successfully, the statement is true for all positive integers.

Important Q/A

Q: What are the two steps of mathematical induction?

A: Basis Step and Inductive Step.

Q: What is the Inductive Hypothesis?

A: It is the assumption that the statement is true for (n=k).

Q: Why is the Basis Step necessary?

A: It verifies that the statement is true for the first value.

Lecture 24 – Mathematical Induction for Divisibility & Sequences

Important Exam Points

Mathematical induction is used to prove divisibility statements.

Induction can prove inequalities.

Recursive sequences can be converted into explicit formulas using induction.

For the sequence ($a_1=2,\;a_n=5a_{n-1}$),

$$
a_n=2\cdot5^{n-1}
$$

For the sequence

$$
d_n=\frac2{n!}
$$

induction proves the explicit formula.

Bernoulli's inequality

$$
1+nx\le(1+x)^n
$$

is proved using induction.

Important Q/A

Q: What is the explicit formula for ($a_n=5a_{n-1}$) with ($a_1=2$)?

A: ($a_n=2\cdot5^{n-1}$).

Q: Which inequality is commonly proved by induction?

A: Bernoulli's Inequality.

Q: Can induction prove recursive sequences?

A: Yes.

Lecture 25 – Methods of Proof

Important Exam Points

A proof is a logical argument that establishes mathematical truth.

Main proof methods are Direct Proof, Contradiction, Contraposition and Counterexample.

A Direct Proof starts from the hypothesis.

Definitions of even, odd, prime, composite, rational and divisibility are frequently used.

An even integer has the form

[
2k
]

An odd integer has the form

[
2k+1
]

A rational number has the form

[
\frac ab,;b\ne0
]

Divisibility means

[
a|b\iff b=ak
]

for some integer (k).

Important Q/A

Q: What is Direct Proof?

A: A proof that begins with the hypothesis and logically reaches the conclusion.

Q: Define an even integer.

A: An integer of the form (2k).

Q: Define divisibility.

A: (a|b) if (b=ak) for some integer (k).

Lecture 26 – Proof by Contradiction

Important Exam Points

Proof by contradiction assumes the statement is false.

The false assumption leads to a contradiction.

The contradiction proves the original statement is true.

The irrationality of (\sqrt2) is proved by contradiction.

There is no greatest integer.

The sum of a rational number and an irrational number is irrational.

There are infinitely many prime numbers.

Important Q/A

Q: What is the first step in contradiction proof?

A: Assume the statement is false.

Q: Which famous theorem is proved by contradiction?

A: (\sqrt2) is irrational.

Q: Why is there no greatest integer?

A: Because if (N) is greatest, then (N+1) is a larger integer.

Lecture 27 – Proof by Contraposition

Important Exam Points

The contrapositive of

[
p\rightarrow q
]

is

[
\neg q\rightarrow\neg p
]

A statement and its contrapositive are logically equivalent.

Contraposition is useful when direct proof is difficult.

"If (n^2) is even, then (n) is even" is commonly proved using contraposition.

"If (n) is divisible by 5, then (n^2) is divisible by 25" is another important application.

Important Q/A

Q: What is the contrapositive of (p\rightarrow q)?

A: (\neg q\rightarrow\neg p).

Q: Why is contraposition used?

A: Because it is sometimes easier than direct proof.

Q: Are a statement and its contrapositive logically equivalent?

A: Yes.

Yes. I'll continue **Lecture 29 to Lecture 45 one by one** in the same exam pattern.

Lecture 29 – Recurrence Relations

Important Exam Points

A recurrence relation defines each term of a sequence using one or more previous terms.

A recurrence relation consists of:
Initial condition and Recursive formula.

A recurrence relation may have a unique solution if sufficient initial conditions are given.

Linear recurrence relations are commonly used in discrete mathematics.

Homogeneous recurrence relations have all terms involving the sequence.

Non-homogeneous recurrence relations contain an additional independent function.

The order of a recurrence relation is determined by the number of previous terms involved.

Recursive formulas can often be converted into explicit formulas.

Recurrence relations are widely used in algorithms and computer science.

Important Q/A

Q: What is a recurrence relation?

A: A formula that defines each term of a sequence using previous terms.

Q: What are the two main parts of a recurrence relation?

A: Initial condition and recursive formula.

Q: What determines the order of a recurrence relation?

A: The largest previous term used.

Q: What is a homogeneous recurrence relation?

A: A relation containing only sequence terms.

Q: What is a non-homogeneous recurrence relation?

A: A recurrence relation containing an additional function besides sequence terms.

MTH202 Important Exam Points with Q/A (Lectures 30–37)

Lecture 30 – Solving Linear Recurrence Relations

Important Exam Points

A linear recurrence relation can be solved by finding its characteristic equation.

The characteristic equation is obtained by replacing (a_n=r^n).

The roots of the characteristic equation determine the general solution.

Distinct roots produce separate exponential terms.

Repeated roots require multiplication by powers of (n).

Initial conditions are used to determine unknown constants.

The explicit formula satisfies both the recurrence relation and the initial conditions.

Important Q/A

Q: What is the first step in solving a linear recurrence relation?

A: Form the characteristic equation.

Q: Why are initial conditions necessary?

A: To determine the unknown constants.

Q: What happens when the characteristic equation has repeated roots?

A: The solution includes powers of (n).

Lecture 31 – Second-Order Linear Recurrence Relations

Important Exam Points

A second-order recurrence relation depends on the previous two terms.

General form:

[
a_n=c_1a_{n-1}+c_2a_{n-2}
]

The characteristic equation is quadratic.

The nature of roots determines the solution.

Distinct roots give two exponential terms.

Equal roots introduce an (n) factor.

Complex roots produce trigonometric solutions.

Important Q/A

Q: What is a second-order recurrence relation?

A: One that depends on the previous two terms.

Q: Which equation is solved first?

A: The characteristic equation.

Q: What type of equation is the characteristic equation?

A: Quadratic.

Lecture 32 – Generating Functions

Important Exam Points

Generating functions transform sequences into power series.

They simplify solving recurrence relations.

The coefficient of each power represents a sequence term.

Generating functions are useful in counting problems.

Ordinary generating functions are the most common.

Power series representation makes algebraic manipulation easier.

Important Q/A

Q: What is a generating function?

A: A power series representing a sequence.

Q: Why are generating functions useful?

A: They simplify recurrence relations and counting problems.

Q: What does the coefficient of (x^n) represent?

A: The (n^{th}) term of the sequence.

Lecture 33 – Counting Principles

Important Exam Points

The Addition Principle is used when choices are mutually exclusive.

The Multiplication Principle is used for successive independent choices.

Factorial notation is fundamental in counting.

[
n!=n(n-1)\cdots2\cdot1
]

By definition,

[
0!=1
]

Counting avoids listing all possibilities.

Many combinatorial problems use multiplication before permutations.

Important Q/A

Q: When is the Addition Principle used?

A: When events are mutually exclusive.

Q: When is the Multiplication Principle used?

A: For successive independent choices.

Q: What is (0!)?

A: 1.

MTH202 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Lecture 34 – Permutations

Important Exam Points

Permutation means arrangement where order matters.

Formula:

$$
^nP_r=\frac{n!}{(n-r)!}
$$

Objects must not repeat unless repetition is allowed.

Permutations of all objects equal ($n!$).

Circular permutations differ from linear permutations.

Important Q/A

Q: What is a permutation?

A: An arrangement in which order matters.

Q: Formula for permutations?

A: ($\displaystyle ^nP_r=\frac{n!}{(n-r)!}$)

Q: Number of arrangements of (n) distinct objects?

A: ($n!$)

Lecture 35 – Combinations

Important Exam Points

Combination means selection where order does not matter.

Formula:

$$
^nC_r=\frac{n!}{r!(n-r)!}
$$

Every combination corresponds to several permutations.

Relationship:

$$
^nP_r = {}^nC_r \times r!
$$

Useful in probability and counting.

Important Q/A

Q: What is a combination?

A: A selection in which order does not matter.

Q: Formula for combinations?

A: $\displaystyle ^nC_r=\frac{n!}{r!(n-r)!}$

Q: Difference between permutation and combination?

A: Order matters in permutations but not in combinations.

Lecture 36 – Binomial Theorem

Important Exam Points

The Binomial Theorem expands powers of binomials.

General expansion:

$$
(a+b)^n=\sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r
$$

Binomial coefficients come from combinations.

The coefficients are symmetric.

Pascal's Triangle generates binomial coefficients.

Middle term depends on whether n is even or odd.

Important Q/A

Q: Which theorem expands $(a+b)^n$?

A: Binomial Theorem.

Q: How are binomial coefficients calculated?

A: Using combinations.

Q: Which triangle gives binomial coefficients?

A: Pascal's Triangle.

Lecture 37 – Binomial Coefficients

Important Exam Points

Binomial coefficients satisfy

$$\binom{n}{r}=\binom{n}{n-r}$$

Pascal's Identity:

$$\binom{n}{r}=\binom{n-1}{r-1}+\binom{n-1}{r}$$

The sum of coefficients of $(a+b)^n$ equals

$$2^n$$

Alternating sum of coefficients equals zero for $n>0$.

Binomial coefficients appear in probability, algebra and combinatorics.

Important Q/A

Q: State Pascal's Identity.

A: $\displaystyle \binom{n}{r}=\binom{n-1}{r-1}+\binom{n-1}{r}$

Q: What is the sum of binomial coefficients?

A: 2^n.

Q: What is the symmetry property of combinations?

A: $\displaystyle \binom{n}{r}=\binom{n}{n-r}$.

MTH202 Important Exam Points with Q/A (Lectures 38–45)

Lecture 38 – Probability

Important Exam Points

Probability measures the likelihood of an event occurring.

The probability of an event is

$$P(E)=\frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

The probability of any event lies between 0 and 1.

An impossible event has probability 0.

A certain event has probability 1.

The complement rule is

$$P(E')=1-P(E)$$

Mutually exclusive events cannot occur simultaneously.

Independent events do not affect each other's probabilities.

Important Q/A

Q: What is the probability of a certain event?

A: 1.

Q: What is the probability of an impossible event?

A: 0.

Q: State the complement rule.

A: ($P(E')=1-P(E)$).

Q: What is the range of probability?

A: ($0\le P(E)\le1$).

Lecture 39 – Conditional Probability and Bayes' Theorem

Important Exam Points

Conditional probability measures the probability of one event given another has occurred.

The conditional probability formula is

[
P(A|B)=\frac{P(A\cap B)}{P(B)}
]

Two events are independent if

[
P(A\cap B)=P(A)P(B)
]

Bayes' Theorem is used to revise probabilities based on new information.

The Law of Total Probability is frequently used with Bayes' Theorem.

Important Q/A

Q: Define conditional probability.

A: The probability of event (A) given that event (B) has occurred.

Q: State the conditional probability formula.

A: (\displaystyle P(A|B)=\frac{P(A\cap B)}{P(B)}).

Q: When are two events independent?

A: When (P(A\cap B)=P(A)P(B)).

Q: Why is Bayes' Theorem important?

A: It updates probabilities using new information.

Lecture 40 – Inclusion–Exclusion Principle

Important Exam Points

The Inclusion–Exclusion Principle avoids double counting.

For two sets,

[
n(A\cup B)=n(A)+n(B)-n(A\cap B)
]

For three sets,

[
n(A\cup B\cup C)=n(A)+n(B)+n(C)-n(A\cap B)-n(B\cap C)-n(A\cap C)+n(A\cap B\cap C)
]

It is widely used in counting and probability.

Important Q/A

Q: Why is the Inclusion–Exclusion Principle used?

A: To avoid double counting.

Q: Write the formula for two sets.

A: (n(A\cup B)=n(A)+n(B)-n(A\cap B)).

Q: Where is this principle commonly applied?

A: Counting and probability.

Lecture 41 – Graph Theory Basics

Important Exam Points

A graph consists of vertices and edges.

Vertices are also called nodes.

Edges connect pairs of vertices.

The degree of a vertex is the number of incident edges.

The sum of all vertex degrees equals twice the number of edges.

A graph with no cycles is called an acyclic graph.

Connected graphs contain a path between every pair of vertices.

Important Q/A

Q: What is a graph?

A: A collection of vertices connected by edges.

Q: What is the degree of a vertex?

A: The number of edges incident to it.

Q: State the Handshaking Lemma.

A: The sum of all vertex degrees equals twice the number of edges.

MTH202 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Lecture 42 – Trees

Important Exam Points

A tree is a connected graph with no cycles.

Every tree with (n) vertices has

[
n-1
]

edges.

There is exactly one simple path between every pair of vertices.

Removing one edge disconnects a tree.

Adding one edge creates exactly one cycle.

Binary trees are important in computer science.

Important Q/A

Q: Define a tree.

A: A connected graph with no cycles.

Q: How many edges does a tree with (n) vertices have?

A: (n-1).

Q: What happens if one edge is added to a tree?

A: Exactly one cycle is formed.

Lecture 43 – Euler and Hamilton Graphs

Important Exam Points

An Euler circuit uses every edge exactly once.

A connected graph has an Euler circuit if every vertex has even degree.

An Euler path uses every edge exactly once but need not end at the starting vertex.

A Hamilton cycle visits every vertex exactly once.

Hamilton paths visit every vertex exactly once.

Euler graphs concern edges, while Hamilton graphs concern vertices.

Important Q/A

Q: What is an Euler circuit?

A: A circuit that traverses every edge exactly once.

Q: What condition guarantees an Euler circuit?

A: Every vertex has even degree.

Q: What is a Hamilton cycle?

A: A cycle that visits every vertex exactly once.

Q: What is the main difference between Euler and Hamilton graphs?

A: Euler concerns edges; Hamilton concerns vertices.

Lecture 44 – Planar Graphs

Important Exam Points

A planar graph can be drawn without crossing edges.

Euler's Formula is

[
V-E+F=2
]

where (V) is vertices, (E) is edges and (F) is faces.

Complete graph (K_5) is non-planar.

Complete bipartite graph (K_{3,3}) is non-planar.

Planarity is important in circuit design and networks.

Important Q/A

Q: What is a planar graph?

A: A graph that can be drawn without edge crossings.

Q: State Euler's Formula.

A: (V-E+F=2).

Q: Name two famous non-planar graphs.

A: (K_5) and ($K_{3,3}$).

Lecture 45 – Graph Coloring and Final Revision

Important Exam Points

Graph coloring assigns colors to vertices so that adjacent vertices have different colors.

The minimum number of colors required is called the chromatic number.

Bipartite graphs require only two colors.

Complete graph (K_n) requires (n) colors.

Graph coloring has applications in scheduling, map coloring and compiler design.

Review all major proof techniques from Lectures 23–45.

Revise mathematical induction thoroughly.

Memorize important formulas for permutations, combinations and probability.

Practice recurrence relation problems.

Understand graph theory definitions and theorems.

Important Q/A

Q: What is graph coloring?

A: Assigning colors to vertices so adjacent vertices have different colors.

Q: What is the chromatic number?

A: The minimum number of colors needed to color a graph.

Q: How many colors are required for a complete graph (K_n)?

A: (n) colors.

Q: How many colors are required for a bipartite graph?

A: Two colors.

Q: Which topics are most important for the final exam from Lectures 23–45?

MTH202 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A: Mathematical Induction, Proof Techniques, Recurrence Relations, Counting Principles, Permutations, Combinations, Binomial Theorem, Probability, Inclusion–Exclusion Principle, Graph Theory, Trees, Euler & Hamilton Graphs, Planar Graphs, and Graph Coloring.

Lectures 28–45 (Core Exam Concepts)

Important Exam Points

Know the difference between Direct Proof, Contradiction, Contraposition and Counterexample.

Understand recursive and explicit sequence formulas.

Master mathematical induction for sums, inequalities and divisibility.

Remember important divisibility properties:

If (a|b) and (b|c), then (a|c).

If (a|b) and (a|c), then (a|(b+c)).

The sum of three consecutive integers is divisible by 3.

The product of an even and an odd integer is even.

The sum of two odd integers is even.

The square of an even integer is even.

If (n) is odd, then (n^3+n) is even.

The sum of rational numbers is rational.

The sum of a rational and an irrational number is irrational.

One counterexample is sufficient to disprove a universal statement.

The set of prime numbers is infinite.

Important Q/A

Q: Which proof method starts by assuming the statement is false?

A: Proof by Contradiction.

Q: Which proof method assumes the statement is true for (n=k)?

A: Mathematical Induction.

Q: Which proof method uses a false example to disprove a statement?

A: Counterexample.

Q: If (a|b) and (b|c), what follows?

A: (a|c).

Q: If (a|b) and (a|c), what follows?

A: (a|(b+c)).

Q: What is the form of every even integer?

A: (2k).

Q: What is the form of every odd integer?

A: (2k+1).

Q: Is the sum of two rational numbers rational?

A: Yes.

Q: Is the sum of a rational and an irrational number rational?

A: No, it is irrational.

Q: How many counterexamples are needed to disprove a universal statement?

A: Only one.

Q: Which theorem proves there are infinitely many prime numbers?

A: Euclid's Theorem.

Q: What are the four major proof techniques in these lectures?

A: Mathematical Induction, Direct Proof, Proof by Contradiction, and Proof by Contraposition.

Example 1

Prove by mathematical induction that

[
1+2+\cdots+n=\frac{n(n+1)}2
]

Solution

Let

$$
P(n):1+2+\cdots+n=\frac{n(n+1)}2
$$

Basis Step

For (n=1),

LHS = 1

RHS

$$
=\frac{1(1+1)}2=1
$$

Hence true.

Inductive Step

Assume

$$
1+2+\cdots+k=\frac{k(k+1)}2
$$

Now,

$$
1+2+\cdots+k+(k+1)
$$

$$
=\frac{k(k+1)}2+(k+1)
$$

$$
=(k+1)\left(\frac{k}{2}+1\right)
$$

$$
=\frac{(k+1)(k+2)}2
$$

which equals RHS.

Hence true for (k+1).

Therefore,

$$
1+2+\cdots+n=\frac{n(n+1)}2
$$

is true for all positive integers.

Example 2

Prove

[
1+3+5+\cdots+(2n-1)=n^2
]

Solution

Let

[
P(n):1+3+\cdots+(2n-1)=n^2
]

Basis Step

For (n=1),

LHS =1

RHS =1

True.

Inductive Step

Assume

[
1+3+\cdots+(2k-1)=k^2
]

Then

[
1+3+\cdots+(2k-1)+(2k+1)
]

[
=k^2+2k+1
]

[
=(k+1)^2
]

Hence true.

Therefore,

$$
1+3+\cdots+(2n-1)=n^2.
$$

Example 3

Prove

$$
1+2+2^2+\cdots+2^n=2^{n+1}-1
$$

Solution

Let

$$
P(n):1+2+\cdots+2^n=2^{n+1}-1
$$

Basis Step

For $(n=0)$,

LHS =1

RHS

$$
=2^1-1=1
$$

True.

Inductive Step

Assume

$$
1+2+\cdots+2^k=2^{k+1}-1
$$

Then

$$
1+2+\cdots+2^k+2^{k+1}
$$

$$
=(2^{k+1}-1)+2^{k+1}
$$

$$
=2^{k+2}-1
$$

Hence true.

Example 4

Prove

$$
1^2+2^2+\cdots+n^2=\frac{n(n+1)(2n+1)}6
$$

Solution

Let

$$
P(n):1^2+\cdots+n^2=\frac{n(n+1)(2n+1)}6
$$

Basis Step

For $(n=1)$,

LHS $=1$

RHS

$$
=\frac{1(2)(3)}6=1
$$

True.

Inductive Step

Assume

$$
1^2+\cdots+k^2=\frac{k(k+1)(2k+1)}6
$$

Then

$$
1^2+\cdots+(k+1)^2
$$

$$
=\frac{k(k+1)(2k+1)}6+(k+1)^2
$$

$$
=\frac{(k+1)(k+2)(2k+3)}6
$$

Hence proved.

Example 5

Prove

$$
1^3+2^3+\cdots+n^3=\left(\frac{n(n+1)}2\right)^2
$$

Solution

Basis Step

For (n=1),

LHS =1

RHS =1

True.

Inductive Step

Assume

$$
1^3+\cdots+k^3=\left(\frac{k(k+1)}2\right)^2
$$

Then

$$
1^3+\cdots+(k+1)^3
$$

$$
=\left(\frac{k(k+1)}2\right)^2+(k+1)^3
$$

Factor and simplify,

$$
=\frac{(k+1)^2(k+2)^2}{4}
$$

$$
=\left(\frac{(k+1)(k+2)}2\right)^2
$$

Hence true.

Example 6

Prove

$$
2+4+6+\cdots+2n=n(n+1)
$$

Solution

Basis Step

For $(n=1)$,

LHS =2

RHS =2

True.

Inductive Step

Assume

$$
2+4+\cdots+2k=k(k+1)
$$

Then

$$
2+4+\cdots+2k+2(k+1)
$$

[
=k(k+1)+2(k+1)
]

[
=(k+1)(k+2)
]

Hence true.

Example 7

Prove

[
3+6+9+\cdots+3n=\frac{3n(n+1)}2
]

Solution

Basis Step

For (n=1),

LHS =3

RHS =3

True.

Inductive Step

Assume true for (k).

Then

[
\frac{3k(k+1)}2+3(k+1)
]

[
=\frac{3(k+1)(k+2)}2
]

Hence proved.

Example 8

Prove

$$
5+10+15+\cdots+5n=\frac{5n(n+1)}2
$$

Solution

Basis Step:

For (n=1),

Both sides =5.

Assume true for (k).

Then

$$
\frac{5k(k+1)}2+5(k+1)
$$

$$
=\frac{5(k+1)(k+2)}2
$$

Hence true.

Example 9

Prove

$$
1+4+7+\cdots+(3n-2)=\frac{n(3n-1)}2
$$

Solution

Basis Step

For (n=1),

LHS =1

RHS =1

True.

Inductive Step

Assume true.

Add next term

[
3(k+1)-2=3k+1
]

Then

[
\frac{k(3k-1)}2+3k+1
]

[
=\frac{(k+1)(3k+2)}2
]

[
=\frac{(k+1)(3(k+1)-1)}2
]

Hence proved.

Example 10

Prove

[
1+6+11+\cdots+(5n-4)=\frac{n(5n-3)}2
]

Solution

Basis Step

For (n=1),

LHS =1

RHS =1

True.

Inductive Step

Assume

$$
1+6+\cdots+(5k-4)=\frac{k(5k-3)}2
$$

Add next term

$$
5(k+1)-4=5k+1
$$

Then

$$
\frac{k(5k-3)}2+5k+1
$$

$$
=\frac{(k+1)(5k+2)}2
$$

$$
=\frac{(k+1)(5(k+1)-3)}2
$$

Hence the statement is true for all positive integers.

PART 1B — Solved Examples 11–20 (Lecture 23: Mathematical Induction)

Example 11

Prove that

$$
\sum_{i=1}^{n}\frac{1}{i(i+1)}=\frac{n}{n+1}
$$

Solution

Let

$$
P(n):\sum_{i=1}^{n}\frac{1}{i(i+1)}=\frac{n}{n+1}
$$

Basis Step

For $(n=1)$,

LHS

$$
=\frac{1}{1\cdot2}=\frac12
$$

RHS

$$
=\frac12
$$

Hence true.

Inductive Step

Assume

$$
\sum_{i=1}^{k}\frac1{i(i+1)}=\frac{k}{k+1}
$$

Now,

$$
\sum_{i=1}^{k+1}\frac1{i(i+1)}
$$

$$
=\frac{k}{k+1}+\frac1{(k+1)(k+2)}
$$

Taking LCM,

$$
=\frac{k(k+2)+1}{(k+1)(k+2)}
$$

$$
=\frac{k^2+2k+1}{(k+1)(k+2)}
$$

$$
=\frac{(k+1)^2}{(k+1)(k+2)}
$$

$$
=\frac{k+1}{k+2}
$$

Hence proved.

Example 12

Prove that

$$\sum_{i=1}^{n}\frac2{i(i+1)}=\frac{2n}{n+1}$$

Solution

Basis Step

For $(n=1)$,

LHS

$$=\frac22=1$$

RHS

$$=\frac21=1$$

True.

Inductive Step

Assume

$$\sum_{i=1}^{k}\frac2{i(i+1)} =\frac{2k}{k+1}$$

Then

$$=\frac{2k}{k+1}+\frac2{(k+1)(k+2)}$$

$$=\frac{2k(k+2)+2}{(k+1)(k+2)}$$

$$=\frac{2(k+1)^2}{(k+1)(k+2)}$$

$$=\frac{2(k+1)}{k+2}$$

Hence true.

Example 13

Prove that

$$1+3+3^2+\cdots+3^n=\frac{3^{n+1}-1}{2}$$

Solution

Basis Step

For $(n=0)$,

LHS

$$=1$$

RHS

$$=\frac31-\frac12=1$$

True.

Inductive Step

Assume

$$1+3+\cdots+3^k=\frac{3^{k+1}-1}{2}$$

Then

$$=\frac{3^{k+1}-1}{2}+3^{k+1}$$

$$=\frac{3^{k+2}-1}{2}$$

Hence proved.

Example 14

Prove that

$$1+5+5^2+\cdots+5^n=\frac{5^{n+1}-1}{4}$$

Solution

Basis Step

For $(n=0)$,

Both sides equal 1.

Inductive Step

Assume

$$1+5+\cdots+5^k=\frac{5^{k+1}-1}{4}$$

Then

$$=\frac{5^{k+1}-1}{4}+5^{k+1}$$

$$=\frac{5^{k+2}-1}{4}$$

Hence true.

Example 15

Prove that

$$2^n\ge n+1$$

for every positive integer.

Solution

Basis Step

For (n=1),

[
2=2
]

True.

Inductive Step

Assume

[
2^k\ge k+1
]

Then

[
2^{k+1}=2(2^k)
]

[
\ge2(k+1)
]

Since

[
2k+2\ge k+2
]

for every positive integer,

therefore

[
2^{k+1}\ge k+2
]

Hence proved.

Example 16

Prove that

[
3^n>2n+1
]

for every (n\ge1).

Solution

Basis Step

For (n=1),

[
3>3
]

Equality holds.

For (n=2),

[
9>5
]

True.

Inductive Step

Assume

[
3^k>2k+1
]

Then

[
3^{k+1}=3\cdot3^k
]

[

3(2k+1)
]

[
=6k+3
]

Since

[
$6k+3>2k+3$
]

therefore

[
$3^{k+1}>2(k+1)+1$
]

Hence proved.

Example 17

Prove that

[
$n<2^n$
]

for all positive integers.

Solution

Basis Step

For $(n=1)$,

[
$1<2$
]

True.

Inductive Step

Assume

[
$k<2^k$
]

Then

[
$k+1\le2k$
]

and

[
$2k<2^{k+1}$
]

Hence

[
$k+1<2^{k+1}$
]

Therefore proved.

Example 18

Prove that

[
$n^2<2^n$
]

for every ($n\ge5$).

Solution

Basis Step

For ($n=5$),

[
$25<32$
]

True.

Inductive Step

Assume

[
$k^2<2^k$
]

Then

[
$(k+1)^2=k^2+2k+1$
]

Since

$$
2k+1\le k^2
$$

for ($k\ge3$),

$$
(k+1)^2\le2k^2
$$

Using induction hypothesis,

$$
2k^2<2^{k+1}
$$

Hence

$$
(k+1)^2<2^{k+1}
$$

Therefore proved.

Example 19

Prove that

$$
5^n-5
$$

is divisible by 5.

Solution

Basis Step

For ($n=1$),

$$
5-5=0
$$

which is divisible by 5.

Inductive Step

Assume

[
5^k-5=5m
]

for some integer (m).

Then

[
5^{k+1}-5
]

[
=5\cdot5^k-5
]

[
=5(5^k-1)
]

Hence divisible by 5.

Therefore true.

Example 20

Prove that

[
2^{2n}-1
]

is divisible by 3.

Solution

Basis Step

For (n=1),

[
4-1=3
]

which is divisible by 3.

Inductive Step

Assume

[
2^{2k}-1=3m
]

Then

[
2^{2(k+1)}-1
]

[
=4\cdot2^{2k}-1
]

[
=4(2^{2k}-1)+3
]

The first term is divisible by 3 by the induction hypothesis, and 3 is divisible by 3.

Hence

[
2^{2(k+1)}-1
]

is divisible by 3.

Therefore the statement is true for all positive integers.

PART 1C — Solved Examples 21–30 (Lectures 24–25)

Example 21

Prove that (n^3-n) is divisible by 3.

Solution

Basis Step

For (n=1),

[
1^3-1=0
]

which is divisible by 3.

Inductive Step

Assume

[
k^3-k=3m
]

for some integer (m).

Now,

[
(k+1)^3-(k+1)
]

[
=k^3+3k^2+3k+1-k-1
]

[
=(k^3-k)+3k^2+3k
]

[
=3m+3(k^2+k)
]

[
=3(m+k^2+k)
]

Hence divisible by 3.

Therefore,

[
n^3-n
]

is divisible by 3 for all positive integers.

Example 22

Prove that (n^3-n) is divisible by 6.

Solution

Basis Step

For (n=1),

[
1^3-1=0
]

which is divisible by 6.

Inductive Step

Assume

[
k^3-k=6m
]

Then

[
(k+1)^3-(k+1)
]

[
=(k^3-k)+3k(k+1)
]

Since one of (k) or (k+1) is even,

[
3k(k+1)
]

is divisible by 6.

Also,

[
k^3-k
]

is divisible by 6 by assumption.

Hence

[
(k+1)^3-(k+1)
]

is divisible by 6.

Therefore proved.

Example 23

Prove that (4^n-1) is divisible by 3.

Solution

Basis Step

For (n=1),

[
4-1=3
]

which is divisible by 3.

Inductive Step

Assume

[
4^k-1=3m
]

Then

[
4^{k+1}-1
]

[
=4(4^k)-1
]

[
=4(4^k-1)+3
]

Both terms are divisible by 3.

Hence

[
4^{k+1}-1
]

is divisible by 3.

Example 24

Prove that (7^n-7) is divisible by 7.

Solution

Basis Step

For (n=1),

[
7-7=0
]

which is divisible by 7.

Inductive Step

Assume

[
7^k-7=7m
]

Then

[
7^{k+1}-7
]

[
=7\cdot7^k-7
]

[
=7(7^k-1)
]

Hence divisible by 7.

Therefore true.

Example 25

Prove that (8^n-2^n) is divisible by 6.

Solution

Basis Step

For (n=1),

[
$8-2=6$
]

which is divisible by 6.

Inductive Step

Assume

[
$8^k-2^k=6m$
]

Then

[
$8^{k+1}-2^{k+1}$
]

[
$=8\cdot8^k-2\cdot2^k$
]

[
$=8(8^k-2^k)+6(2^k)$
]

Each term is divisible by 6.

Hence proved.

Example 26

Prove that the sum of two odd integers is even.

Solution

Let

[
$m=2a+1,\qquad n=2b+1$
]

where ($a,b\in\mathbb Z$).

Then

[
$m+n$
]

[
=(2a+1)+(2b+1)
]

[
=2(a+b+1)
]

Since (a+b+1) is an integer,

[
m+n
]

is even.

Hence proved.

Example 27

Prove that the product of an even integer and an odd integer is even.

Solution

Let

[
m=2a
]

and

[
n=2b+1
]

Then

[
mn
]

[
=2a(2b+1)
]

[
=2[a(2b+1)]
]

Hence

[
mn
]

is even.

Example 28

Prove that the square of an even integer is even.

Solution

Let

[
n=2k
]

Then

[
n^2
]

[
=(2k)^2
]

[
=4k^2
]

[
=2(2k^2)
]

Hence

[
n^2
]

is even.

Example 29

Prove that if (n) is odd, then (n^3+n) is even.

Solution

Let

[
$n=2k+1$
]

Then

[
n^3+n
]

[
$=(2k+1)^3+(2k+1)$
]

[
$=8k^3+12k^2+6k+1+2k+1$
]

[
$=8k^3+12k^2+8k+2$
]

[
$=2(4k^3+6k^2+4k+1)$
]

Hence

[
n^3+n
]

is even.

Example 30

Prove that if the sum of two integers is even, then their difference is also even.

Solution

Suppose

[
$m+n$
]

is even.

Then

[
m+n=2k
]

for some integer (k).

Hence

[
m=2k-n
]

Now,

[
m-n
]

[
=(2k-n)-n
]

[
=2k-2n
]

[
=2(k-n)
]

Since (k-n) is an integer,

[
m-n
]

is even.

Hence proved.

PART 1D — Solved Examples 31–40 (Lecture 25: Direct Proof)

Example 31

Prove that the sum of two rational numbers is rational.

Solution

Suppose

$$
r=\frac{a}{b}, \qquad s=\frac{c}{d}
$$

where $a,b,c,d\in\mathbb Z$ and $b\neq0,\;d\neq0$.

Now,

$$
r+s=\frac{a}{b}+\frac{c}{d}
$$

$$
=\frac{ad+bc}{bd}
$$

Since $ad+bc$ and bd are integers and $bd\neq0$,

$$
r+s
$$

is a rational number.

Hence proved.

Example 32

Prove that the product of two rational numbers is rational.

Solution

Let

$$
r=\frac{a}{b},\qquad s=\frac{c}{d}
$$

where

$$
b\neq0,\qquad d\neq0.
$$

Then

$$
rs
$$

$$
=\frac{a}{b}\times\frac{c}{d}
$$

$$
=\frac{ac}{bd}
$$

Since (ac) and (bd) are integers and $(bd\neq0)$,

the product is rational.

Hence proved.

Example 33

Given two rational numbers $(r<s)$, prove that

$$
x=\frac{r+s}{2}
$$

is also rational and satisfies $(r<x<s)$.

Solution

Since

$$
r=\frac ab,\qquad
s=\frac cd
$$

are rational,

their sum

$$
r+s
$$

is rational.

Also,

$$
2
$$

is a non-zero rational number.

Therefore,

[
\frac{r+s}{2}
]

is rational.

Now,

Since

[
r<s,
]

adding (r) to both sides,

[
2r<r+s
]

Dividing by 2,

[
r<\frac{r+s}{2}
]

Similarly,

adding (s),

[
r+s<2s
]

Dividing by 2,

[
\frac{r+s}{2}<s
]

Hence,

[
r<x<s.
]

Example 34

Prove that if (a|b) and (b|c), then (a|c).

Solution

Suppose

$$a|b,\qquad b|c.$$

Then

$$b=ar$$

for some integer (r),

and

$$c=bs$$

for some integer (s).

Substituting,

$$c=(ar)s$$

$$=a(rs)$$

Since

$$rs$$

is an integer,

$$a|c.$$

Hence proved.

Example 35

Prove that if ($a|b$) and ($a|c$), then ($a|(b+c)$).

Solution

Suppose

$$b=ar,\qquad c=as$$

where (r,s) are integers.

Then

$$b+c$$

$$=ar+as$$

$$=a(r+s)$$

Since

$$r+s$$

is an integer,

$$a|(b+c).$$

Hence proved.

Example 36

Prove that if $(a|b)$ and $(a|c)$, then $(a|(b-c))$.

Solution

Suppose

$$b=ar,\qquad c=as.$$

Then

[
b-c
]

[
=ar-as
]

[
=a(r-s)
]

Since

[
r-s
]

is an integer,

[
a|(b-c).
]

Hence proved.

Example 37

Prove that the sum of three consecutive integers is divisible by 3.

Solution

Let the integers be

[
n,;n+1,;n+2.
]

Their sum is

[
n+(n+1)+(n+2)
]

[
=3n+3
]

[
=3(n+1)
]

Since

[
n+1
]

is an integer,

the sum is divisible by 3.

Hence proved.

Example 38

Prove that the product of two consecutive integers is even.

Solution

Let the integers be

[
n,;n+1.
]

Among two consecutive integers,

one must be even.

Therefore,

their product

[
n(n+1)
]

contains a factor of 2.

Hence

[
n(n+1)
]

is even.

Example 39

Prove that the product of three consecutive integers is divisible by 6.

Solution

Let the integers be

[
n,;n+1,;n+2.
]

Among any three consecutive integers,

one is divisible by 3.

Also,

at least one is even.

Therefore,

their product contains both factors

[
2\times3=6.
]

Hence

[
n(n+1)(n+2)
]

is divisible by 6.

Example 40

Prove that there exists an integer greater than 5 for which

[
2^n-1
]

is prime.

Solution

Take

[
n=7.
]

Then

[
2^7-1
]

[
=128-1
]

[
=127.
]

Since

127 has no positive divisors other than

[
1
]

and

[
127,
]

it is prime.

Hence there exists an integer greater than 5 for which

[
2^n-1
]

is prime.

Hence proved.

PART 1E — Solved Examples 41–50 (Lectures 25–26: Counter Examples & Proof by Contradiction)

Example 41

Disprove the statement:

"For all real numbers (a) and (b), if (a<b), then (a^2<b^2)."

Solution

Take

[
a=-5,\qquad b=-2.
]

Clearly,

[
-5<-2.
]

Now,

[
a^2=(-5)^2=25
]

and

[
b^2=(-2)^2=4.
]

Since

[
25>4,
]

the conclusion

[
a^2<b^2
]

is false.

Hence the given statement is **disproved**.

Example 42

Disprove the statement:

"For every integer (n), (n^2+n+17) is prime."

Solution

Take

[
n=16.
]

Then

[
n^2+n+17
]

[
=16^2+16+17
]

[
=256+16+17
]

[
=289
]

[
=17\times17.
]

Since 289 is not prime,

the statement is false.

Hence disproved.

Example 43

Disprove the statement:

"The product of any two irrational numbers is irrational."

Solution

Take

[
\sqrt2
]

and

[
\sqrt2.
]

Both are irrational.

Their product is

[
\sqrt2\times\sqrt2
]

[
=2,
]

which is rational.

Hence the statement is false.

Example 44

Find a counterexample to:

"For every prime number (p), (p+2) is also prime."

Solution

Take

[
p=7.
]

Then

[
p+2=9.
]

Since

[
9=3\times3,
]

9 is not prime.

Hence

[
p=7
]

is a counterexample.

Example 45

Prove by contradiction that there is no greatest integer.

Solution

Assume there exists a greatest integer (N).

Then every integer satisfies

[
n\le N.
]

Consider

[
N+1.
]

Since the sum of two integers is an integer,

[
N+1
]

is also an integer.

Also,

[
N+1>N.
]

This contradicts the assumption that

[
N
]

is the greatest integer.

Hence our assumption is false.

Therefore,

there is **no greatest integer**.

Example 46

Prove by contradiction that if (n^2) is even, then (n) is even.

Solution

Assume

[
n^2
]

is even but

[
n
]

is odd.

Then

[
n=2k+1
]

for some integer (k).

Now,

[
n^2=(2k+1)^2
]

[
=4k^2+4k+1
]

[
=2(2k^2+2k)+1.
]

Thus

[
n^2
]

is odd,

which contradicts the assumption that

[
n^2
]

is even.

Hence

[
n
]

must be even.

Example 47

Prove by contradiction that if (n^3+5) is odd, then (n) is even.

Solution

Assume

[
n^3+5
]

is odd,

but

[
n
]

is odd.

Then

[
n^3
]

is also odd.

Now,

[
(n^3+5)-n^3=5.
]

The difference of two odd numbers is even.

Hence

[
5
]

would be even,

which is impossible.

Therefore,

our assumption is false.

Hence

[
n
]

is even.

Example 48

Prove by contradiction that the sum of a rational number and an irrational number is irrational.

Solution

Suppose

[
r
]

is rational,

[
s
]

is irrational,

and

[
r+s
]

is rational.

Then

[
s=(r+s)-r.
]

Since the difference of two rational numbers is rational,

[
s
]

must be rational.

This contradicts the assumption that

[
s
]

is irrational.

Hence

[
r+s
]

is irrational.

Example 49

Prove by contradiction that (\sqrt2) is irrational.

Solution

Assume

[
\sqrt2
]

is rational.

Then

[
\sqrt2=\frac{m}{n},
]

where

[
m
]

and

[
n
]

have no common factor.

Squaring,

[
2=\frac{m^2}{n^2}.
]

Hence,

[
m^2=2n^2.
]

Therefore,

[
m^2
]

is even,

so

[
m
]

is even.

Let

[
m=2k.
]

Substituting,

[
4k^2=2n^2
]

[
n^2=2k^2.
]

Thus

[
n
]

is also even.

Therefore,

both

[
m
]

and

[
n
]

are divisible by 2,

contradicting the assumption that they have no common factor.

Hence,

[
\sqrt2
]

is irrational.

Example 50

Prove by contradiction that if a prime number (p) divides both (a) and (a+1), then a contradiction occurs.

Solution

Assume

[
p\mid a
]

and

[
p\mid(a+1).
]

Then

[
a=pr
]

and

[
a+1=ps
]

for some integers (r) and (s).

Subtracting,

[
(a+1)-a=ps-pr.
]

Therefore,

[
1=p(s-r).
]

Hence,

[
p\mid1.
]

But the only divisors of 1 are

[
1
]

and

[
-1,
]

whereas a prime number satisfies

[
p>1.
]

This is impossible.

Hence the assumption is false.

Therefore, **no prime number can divide both (a) and (a+1).**

THE END

MTH202 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS